COLLEGE OF BUSINESS EDUCATION
DAR ES SALAAM
TEST 1. A
TFC/TNC OCT 2023/2024 BIT
SUBJECT: DATA STRUCTURE AND ALGORITHMS
PAPER CODE: IT07313

DATE: 16th JANUARY 2026 **TIME: 45:00 MIN**

Instructions

- *This paper consists of* ***two questions on one Page; attempt all.***
- *This test contributes 12.5 marks to the module assessment*

Question One

Define the following terms as they are used in Data structure

i) Class **(1.5 Mark)**

Answer

It has more than one definition, and you can choose either

- Is a blueprint for creating objects
- Is an object constructor
- Is a collection of functions and variables

ii) Object **(1.5 Mark)**

Answer

It has more than one definition, and you can choose either

- Is any real-world entity that can be implemented in to computer program
- It is an instance of a class

iii) dynamic data structure **(1.5 Mark)**

Answer

Is a type of linear data structure where the size of the structure is flexible. This means the structure allows adding more elements at runtime. This is because it allocates memory during runtime

Question Two

With an example, explain how the bubble sort can be used to sort the following array elements in ascending order **int A[]=new int{14,33,27,35,10}** **(8 Marks)**

Answer

An example on how to sort an array **int A[]=new int{14,33,27,35,10}** using bubble sort

1st Iteration ***((02 Marks)***

14, 33, 27, 35, 10,	Compare 14 and 33	14 <33	No Swap
14, 33, 27, 35, 10,	Compare 33 and 27	33 >27	Swap
14, 27, 33, 35, 10,	Compare 33 and 35	33 <35	No Swap
14, 27, 33, 35, 10,	Compare 35 and 10	35 >10	Swap

2nd Iteration ***((02 Mark)***

14, 27, 33, 10, 35,	Compare 14 and 27	14<27	No Swap
14, 27, 33, 10, 35,	Compare 27 and 33	27<33	No Swap
14, 27, 33, 10, 35,	Compare 33 and 10	33>10	Swap
14, 27, 10, 33, 35,	Compare 33 and 35	33<35	No Swap

3rd Iteration ***((02 Mark)***

14, 27, 10, 33, 35,	Compare 14 and 27	14<27	No Swap
14, 27, 10, 33, 35,	Compare 27 and 10	27>10	Swap
14, 10, 27, 33, 35,	Compare 27 and 33	27<33	No Swap
14, 10, 27, 33, 35,	Compare 33 and 35	33<35	No Swap

4th Iteration ***((02 Mark)***

14, 10, 27, 33, 35,	Compare 14 and 10	14>10	Swap
10, 14, 27, 33, 35,	Compare 10 and 14	10<14	No Swap